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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S5 (R, S) / S3 (PT) (R, S) Examination December 2023 (2019 Scheme)



Course Code: EET 307

Course Name: SYNCHRONOUS AND INDUCTION MACHINES

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- 1 The stator winding of an alternator has 48 slots. A 4 pole, three phase winding is made on the stator. Each coil spans 11 slots. Calculate the pitch factor. (3)
- 2 Derive an expression for distribution factor in an alternator. (3)
- 3 Draw the phasor diagram of a cylindrical pole type alternator for (i) lagging power factor load (ii) leading power factor load. (3)
- 4 X_d is greater than X_q in salient pole machine. Justify (3)
- 5 Explain the working of synchronous condenser. (3)
- 6 "Power factor of three induction motor is low at starting but it improves as the machine is loaded". Justify the statement. (3)
- 7 Briefly explain the phenomenon of cogging in three phase induction motor (3)
- 8 Explain the effect of varying the rotor resistance on torque-slip curve of three phase induction motor (3)
- 9 Sketch the torque-slip curve of single phase induction motor (3)
- 10 Mention few application of capacitor start capacitor run induction motor (3)

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- 11 a) Derive the emf equation of an alternator (4)
- b) A 50 Hz, 600 r.p.m, salient pole alternator has a sinusoidal flux density having a maximum value of 1 Wb/m². The alternator has 180 slots wound with 2-layer, 3-turn coils. The coil span is 15 slots and phase spread is 60°. The armature diameter is 1.25 m and core length 0.45 m. find (i) flux/pole (ii) pitch factor (iii) distribution factor (iv) peak value of emf per conductor (v) peak value of emf per coil (vi) r.m.s phase voltage if the machine is star-connected. (10)

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- 12 a) List the advantages of stationary armature type alternators over rotating armature types (6)
- b) A 12 pole , 3 phase, star-connected alternator has 72 slots. The flux per pole is 0.0988 Wb. Calculate : (8)
- the speed of rotation if the frequency of the generated e.m.f is 50 Hz.
 - the terminal e.m.f for full-pitch coils and 8 conductors per slot
 - the terminal e.m.f if the coil span is reduced to 2/3 of the pole pitch.

Module -2

- 13 a) The regulation obtained by EMF method is always greater than actual value. Justify (5)
- b) A 550V, 55 kVA, 1-phase alternator has an effective resistance of 0.2 ohm. A field current of 10 A produces an armature current of 200 A on short circuit and an electromotive force of 450 V on open circuit. Calculate (a) the synchronous impedance and reactance, and (b) the full load regulation with power factor 0.8 lagging. (9)
- 14 a) Explain the advantage of determining voltage regulation by ASA method over other methods. (5)
- b) A 3.5 MVA, Y connected alternator rated at 4160 volts at 50 Hz has open-circuit characteristics given by the following data: (9)

Field current (A)	50	100	150	200	250	300	350	400
Line E.M.F (V)	1620	3150	4160	4750	5130	5370	5550	5650

A field current of 200 A is found necessary to circulate full-load current on short-circuit of the alternator. Calculate by the full load regulation at 0.8 pf lag of alternator by Optimistic method. Neglect armature resistance.

Module -3

- 15 a) Sketch and explain V and inverted V curves of synchronous motor (7)
- b) Estimate the stator current, equivalent rotor current, efficiency, and power factor at a slip of 5% for an three phase induction motor with the following data (7)
- Stator impedance: $(1+j3)$ ohm

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Equivalent rotor impedance at stand still : $(1+j2)$ ohm

No load shunt impedance $(10+j50)$ ohm

Voltage per phase = 250 V.

Use approximate equivalent circuit. Assume mechanical losses to be zero.

- 16 a) Explain the principle of operation of synchronous motor with neat sketch (7)
- b) A three phase induction motor with star connected rotor has an induced electromotive force of 60 V between slip-rings at stand-still on open circuit with normal voltage applied to the stator. The resistance and standstill reactance of each rotor phase are 0.6 ohm and 4 ohm respectively. Calculate the current per phase in the rotor (a) when at stand still and connected to a star connected rheostat of resistance 5 ohm and reactance 2 ohm per phase; (b) when running short circuited with 4% slip.

Module -4

- 17 a) Explain star-delta method of starting of 3 phase induction motor with a neat sketch (7)
- b) A 3 h.p, induction motor with full load efficiency and power factor of 0.83 and 0.8 respectively, has a short circuit current of 3.5 times full load current. Estimate the line current at the instant of starting the motor from a 500 V supply by means of a star-delta switch. Ignore the magnetizing current. (7)
- 18 a) Explain any one method of speed control applicable to three phase induction motor (7)
- b) The cages of a double-cage induction motor have stand still impedances of $3.5+j1.5$ ohm and $0.6+j7.0$ ohm respectively. The full-load slip is 6%. Find the starting torque at normal voltage in terms of full-load torque. Neglect stator impedance and magnetizing current. (7)

Module -5

- 19 a) Draw and explain equivalent circuit of single phase induction motor. (7)
- b) Explain the self excited operation of induction generator. (7)
- 20 a) Explain the working of split phase and capacitor start single phase induction motor (8)
- b) Mention few applications of single phase induction motor (6)
